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Feature & Ticket List IPO Allotment Verification System — Development Backlog Prepared by: Bhavya Bulani Version 1.2 · July 2026

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Introduction

This backlog breaks the IPO Allotment Verification System’s requirements into epics and individual development tickets. Estimates are in story points (Fibonacci-style: 1, 2, 3, 5, 8, 13) and are placeholders for the team to refine.

Version 1.2 adds **EPIC-10**, covering the switch from SQLite to MySQL as the primary database. Because this is an infrastructure change rather than a new feature, EPIC-10 is scheduled early — its tickets are prerequisites for every other epic that reads or writes data, rather than being additive work at the end.

Epic Summary

Epic ID	Epic Name	Related FRs
EPIC-1	Input & Validation	FR-001-FR-006, FR-020

Epic ID	Epic Name	Related FRs
EPIC-2	IPO Selection & Master Data	FR-007
EPIC-3	Registrar Integration & Fallback	FR-008–FR-010
EPIC-4	Parallel Processing Engine	FR-011, FR-012
EPIC-5	Results, Reporting & Dashboard	FR-013–FR-015
EPIC-6	Search & History	FR-016, FR-017
EPIC-7	Notifications & Logging	FR-018, FR-019
EPIC-8	IPO Auto-Detection, Multi-IPO Runs & Result Caching	FR-021–FR-024
EPIC-9	CAPTCHA Handling Strategy	FR-025–FR-026
EPIC-10	MySQL Data Layer	FR-027

EPIC-1 — Input & Validation

Ticket	Description	Priority	Estimate
TICKET-101	Build single-client entry form (PAN/Client Code + IPO selector)	High	3
TICKET-102	Implement PAN format validator	High	2
TICKET-103	Implement broker-specific Client Code validator	High	3
TICKET-104	Build Excel upload endpoint accepting .xlsx/.xls only	High	3
TICKET-105	Implement 10,000-row cap with clear rejection message	High	1
TICKET-106	Build column auto-detection	High	5
TICKET-107	Build validation summary UI	High	3
TICKET-108	Ensure invalid rows are excluded from outbound registrar requests, across all selected IPOs	High	2

EPIC-2 — IPO Selection & Master Data

Ticket	Description	Priority	Estimate
TICKET-201	Integrate IPO master data source (e.g., Upstox IPO API)	High	5
TICKET-202	Build IPO sync job (scheduled/manual refresh)	Medium	3
TICKET-203	Build IPO selector component supporting multi-select	High	3

EPIC-3 — Registrar Integration & Fallback

Ticket	Description	Priority	Estimate
TICKET-301	Define common registrar service interface	High	3
TICKET-302	Implement Link Intime registrar module	High	8
TICKET-303	Implement KFin Technologies registrar module	High	8
TICKET-304	Implement Bigshare registrar module	High	8
TICKET-305	Implement MUFG Intime registrar module	High	8
TICKET-306	Build fallback orchestrator	High	5
TICKET-307	Add per-registrar rate limiting / request pacing	High	3

EPIC-4 — Parallel Processing Engine

Ticket	Description	Priority	Estimate
TICKET-401	Build asyncio-based worker pool for concurrent client checks	High	8
TICKET-402	Make worker pool size configurable	Medium	2
TICKET-403	Build job/queue tracking for bulk runs	High	5

Ticket	Description	Priority	Estimate
TICKET-404	Build progress-reporting endpoint (polling or streaming)	Medium	3
TICKET-405	Build frontend Progress Screen	Medium	3
TICKET-406	Tune MySQL connection pool size against configured worker concurrency; add saturation monitoring	High	3

EPIC-5 — Results, Reporting & Dashboard

Ticket	Description	Priority	Estimate
TICKET-501	Implement result status categorization (6 categories)	High	3
TICKET-502	Build Results Table UI with status badges	High	3
TICKET-503	Build Excel export	High	3
TICKET-504	Build Dashboard summary	Medium	2

EPIC-6 — Search & History

Ticket	Description	Priority	Estimate
TICKET-601	Build client-side search/filter on Results Table	Medium	2
TICKET-602	Build run history storage	Medium	3
TICKET-603	Build History screen	Medium	3

EPIC-7 — Notifications & Logging

Ticket	Description	Priority	Estimate
TICKET-701	Build run-completion notification	Low	2
TICKET-702	Build run-level logging	Medium	3
TICKET-703	Build log viewer for troubleshooting	Low	3

EPIC-8 — IPO Auto-Detection, Multi-IPO Runs & Result Caching

Ticket	Description	Priority	Estimate
TICKET-801	Build IPO Auto-Detection Service	High	5
TICKET-802	Implement auto_detected/validated fields and validation gate	High	3
TICKET-803	Add manual “sync now” trigger	Medium	2
TICKET-804	Add “Auto-detected” badge to IPO entries in the frontend selector	Low	1
TICKET-805	Migrate upload_batches off single ipo_id to the new batch_ipos join table	High	3
TICKET-806	Upgrade IPO selector component to multi-select, with selected-IPO chips	High	5
TICKET-807	Build Multi-IPO Orchestrator	High	8
TICKET-808	Update progress tracking and Progress Screen for aggregate + per-IPO breakdown	Medium	3
TICKET-809	Update Results Table, Dashboard, and Excel export for multi-IPO breakdown and IPO filter	High	5
TICKET-810	Design and implement Result Cache Layer	High	5
TICKET-811	Add served_from_cache / cache_expires_at fields to allotment_results; write-through caching	High	3
TICKET-812	Implement configurable cache TTL and expiry-aware cache lookup	Medium	3

Ticket	Description	Priority	Estimate
TICKET-813	Add “Cached” source indicator to Results Table, Dashboard, and Excel export	Medium	2
TICKET-814	Add cache-hit count to run-level logging and Progress Screen	Medium	2
TICKET-815	Update Security & Access Control safeguards for cache and auto-detection	High	3

EPIC-9 — CAPTCHA Handling Strategy

Ticket	Description	Priority	Estimate
TICKET-901	Define CaptchaProvider interface	High	2
TICKET-902	Build CAPTCHA detection in each registrar module	High	5
TICKET-903	Implement Manual-CaptchaProvider	High	5
TICKET-904	Build frontend CaptchaPrompt component	High	3
TICKET-905	Implement CAPTCHA Manager routing logic	High	5
TICKET-906	Build optional SolverProvider adapter(s)	Low	5
TICKET-907	Record which path produced each result	Medium	2
TICKET-908	Document CAPTCHA-solver compliance considerations	Medium	1

EPIC-10 — MySQL Data Layer (new in Version 1.2)

Ticket	Description	Priority	Estimate
TICKET-1001	Provision MySQL 8.0+ instance for local development (Docker Compose) and document setup for staging/production	High	3
TICKET-1002	Define SQLAlchemy ORM models for all seven tables (clients, ipos, registrars, upload_batches, batch_ipos, allotment_results, run_logs), matching the Database Design Document's MySQL types	High	5
TICKET-1003	Set up Alembic and write the initial migration creating the full schema (InnoDB, utf8mb4)	High	3
TICKET-1004	Implement connection pooling (session factory sized to configured worker concurrency) and credential loading from environment variables	High	3
TICKET-1005	Implement the composite index set from the Database Design Document's Indexing Strategy, and verify query plans for the cache-lookup and history queries	Medium	3
TICKET-1006	Replace any placeholder SQLite-specific code paths from earlier prototyping with the MySQL-targeted data-access layer	High	3

Ticket	Description	Priority	Estimate
TICKET-1007	Write a batched purge/retention job for run_logs and allotment_results (avoiding single unbounded DELETE statements)	Medium	2
TICKET-1008	Add MySQL connection-pool saturation and slow-query monitoring/logging	Medium	2
TICKET-1009	Configure and document a MySQL backup procedure (mysqldump or managed provider backups) with the same access restrictions as the live database	Medium	2
TICKET-1010	Load-test parallel-worker writes against MySQL at target concurrency (10-30 workers) to confirm no lock contention or dropped results	High	3

Backlog — Future Roadmap Items (Not Yet Scheduled)

Ticket	Description	Notes
FUTURE-02	Email / WhatsApp notifications to clients	Requires messaging provider integration
FUTURE-04	User login with role-based access control (admin/operator)	See Security & Access Control Document — planned as the next major enhancement
FUTURE-05	Public REST API for other internal systems	Requires API key management
FUTURE-07	Retry with exponential backoff for transient failures	Refinement of fallback orchestrator
FUTURE-08	Per-user audit logging	Depends on FUTURE-04
FUTURE-10	Distributed/shared cache (e.g., Redis) for multi-instance deployments	Only needed once the application is scaled horizontally
FUTURE-11	MySQL read replicas for reporting/history queries	Only needed if reporting load begins to contend with write-heavy check runs

Note: FUTURE-09 (“Migration from SQLite to PostgreSQL/MySQL”) from the Version 1.1 backlog has been **completed and removed** from the future roadmap — it is now EPIC-10 in active scope, per this Version 1.2 update. FUTURE-01, FUTURE-03, and FUTURE-06 remain retired into EPIC-8 as of Version 1.1. FUTURE-04 (role-based access control) remains explicitly deferred.

Suggested Sprint Grouping (Indicative)

Sprint	Focus
Sprint 0	EPIC-10 (MySQL Data Layer) — provisioning, ORM models, initial migration, connection pooling. This is a prerequisite sprint since every other epic depends on the data layer.
Sprint 1	EPIC-1 (Input & Validation), EPIC-2 (IPO Selection & Master Data)
Sprint 2	EPIC-3 (Registrar Integration & Fallback)
Sprint 3	EPIC-4 (Parallel Processing Engine), including MySQL connection-pool tuning (TICKET-406)
Sprint 4	EPIC-5 (Results, Reporting & Dashboard)
Sprint 5	EPIC-6 (Search & History), EPIC-7 (Notifications & Logging), hardening and QA
Sprint 6	EPIC-8, Part A — IPO Auto-Detection (TICKET-801–804) and Multi-IPO foundation (TICKET-805–807)
Sprint 7	EPIC-8, Part B — Multi-IPO UI/reporting (TICKET-808–809), Result Caching (TICKET-810–815), and regression testing
Sprint 8	EPIC-9 — CAPTCHA Handling Strategy, plus final regression testing
Sprint 9	EPIC-10 remainder — load testing (TICKET-1010), backup procedure (TICKET-1009), monitoring (TICKET-1008), and final MySQL hardening pass before release